



TFlow

for Unity

Motion Vector Baker

TFlow is a motion vector and motion blur generator that helps increase the utility and quality of your flipbooks. With provided shader examples it adapts to many rendering pipelines.

 TFlow Promo

[Get TFlow on the Asset Store](#)

[Version 1.2](#)

[Web demo](#) 💡 The demo showcase the blending result you can get thanks to TFlow. This is not a demo of the baking tool.

[UE documentation](#)

For any question related to TFlow or for technical support, email us at support@tuataravfx.com.

Don't hesitate to send us feature requests as well.

How does it work?

Optical Flow

Use our baking tool to generate motion vectors for your flipbook.

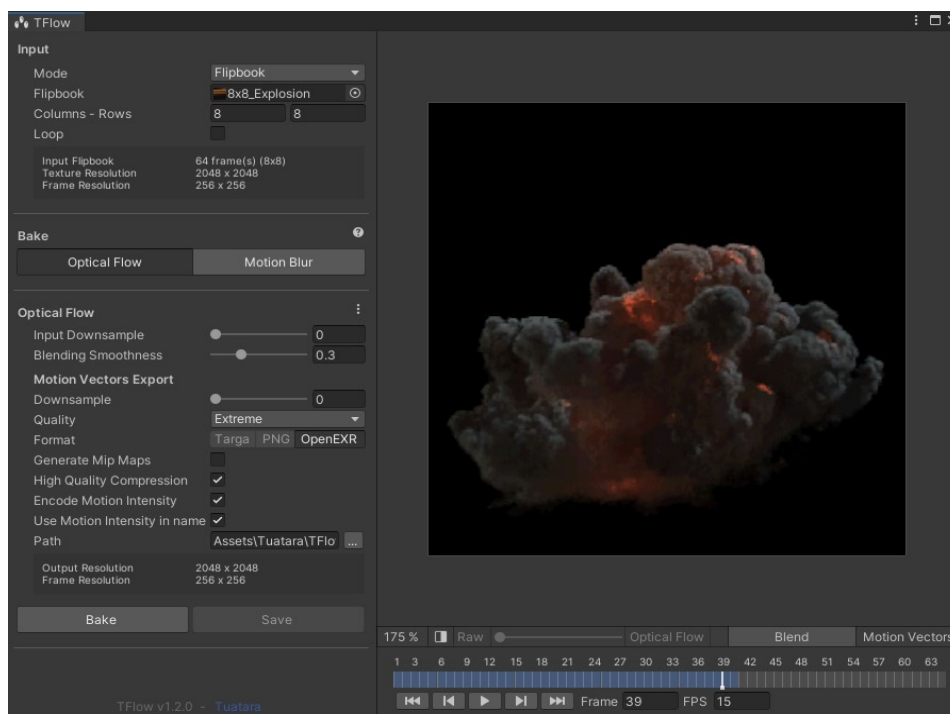
Use the motion vectors texture and the flipbook at runtime to get optical flow blending.

Motion Blur

Bake motion blur directly on your flipbook using our tool.

What's inside

- An editor baking tool for Unity that generates motion vectors and motion blur textures.



- Shaders, nodes and material examples to use optical flow blending in your projects.

Support

TFlow baking tool works on any Windows platform/pipeline.

Unity Version	Baking tool support
Windows Unity 2019.4 LTS	✓
Windows Unity 2020.3 LTS	✓
Windows Unity 2021.1	✓
Windows Unity 2021.2b	✓

A set of shader functions is available to use on any Unity or pipeline version. Using these functions you'll be able to support optical flow in your own shaders. We also provide integration examples for Universal RP, High Definition RP, Shader Graph and Amplify Shader Editor.

Pipeline	Shaders support
Universal RP 10+	✓
High Definition RP 10+	✓
Amplify Shader Editor	✓
VFX Graph	✓
Shader Graph	✓
Custom Shaders	✓
Other or older Unity pipelines	✓ but without examples (see Getting Started > Shader Functions to write your own)

👉 If you need support for a specific Unity or pipeline version, feel free to contact us.

Getting started

Unity Installation

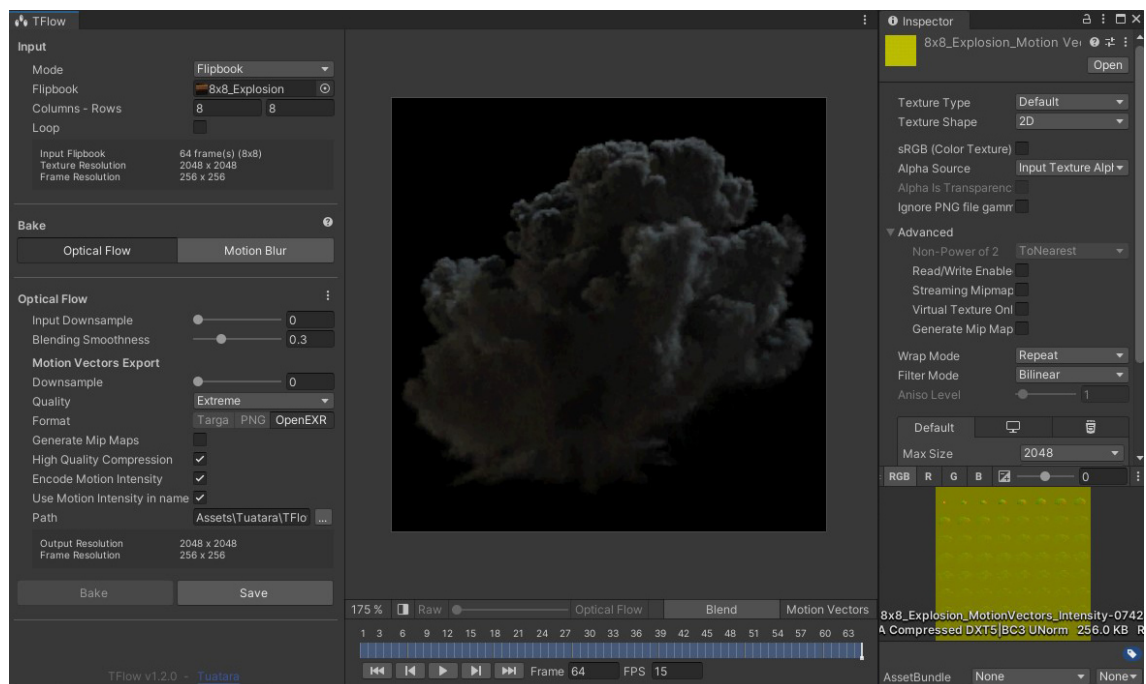
- [Download TFlow package for Unity.](#)
- Import the package in your Unity project

Baking Optical Flow Motion Vectors

- Open the tool **Window > Tuatara > TFlow** or right click on your flipbook **Tuatara > Open TFlow**.
- Drop your flipbook in the texture field and fill in the size
- Check **Loop** if your sequence is supposed to loop
- Choose **Optical Flow** in Baking
- Play with the **Blending Smoothness** and the timeline to preview blending

💡 *The column and row count will be filled automatically if the size is included in the file name like "**COLUMNSxROWS**".*

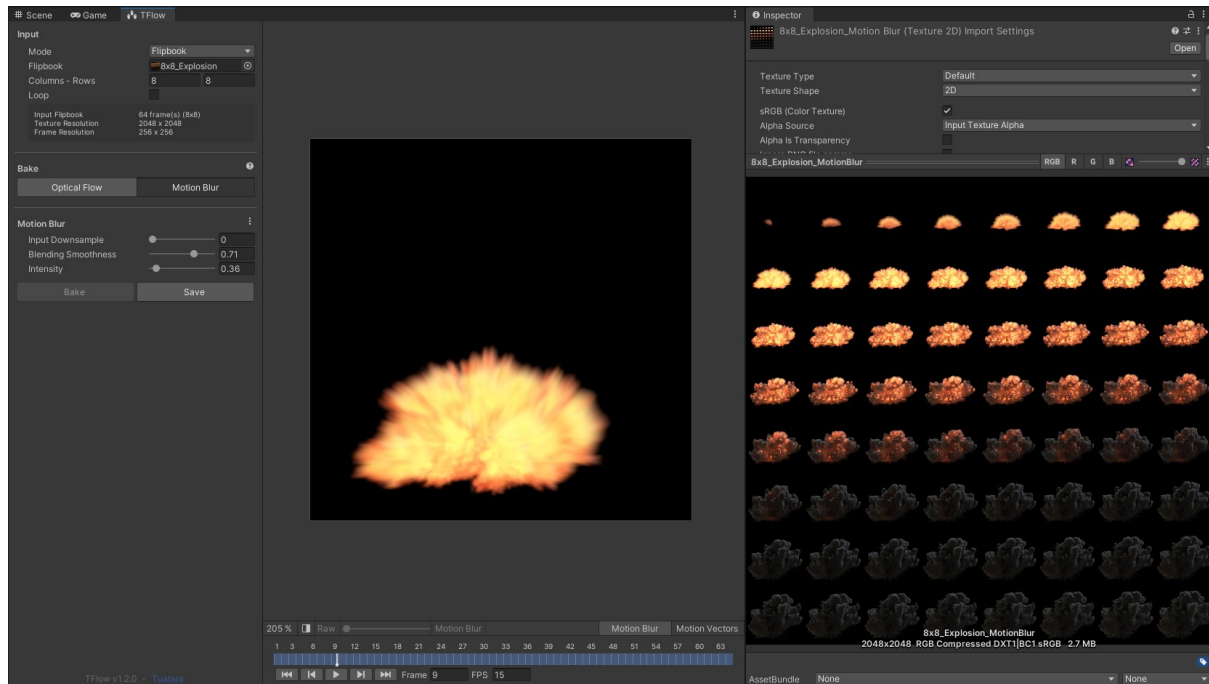
- Press Bake and Save. The motion vectors texture should appear next to your flipbook. You can then use it in your shaders to produce realtime smooth blending, check the Blending section that matches your rendering pipeline down below.



Baking Motion Blur

You can bake motion blur on your flipbooks using TFlow.

- Choose Motion Blur under Baking
- Play with the settings
- Press Bake and Save. The motion blur flipbook will be saved next to the input flipbook.

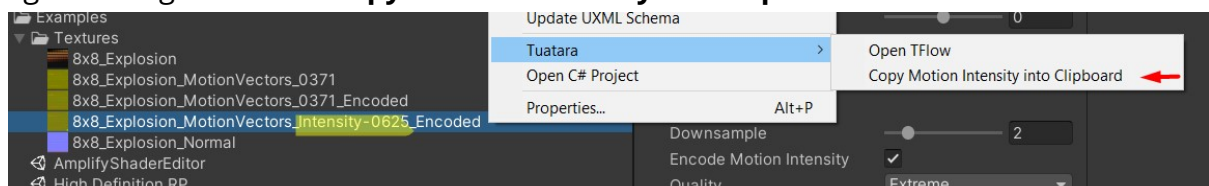


Motion intensity

The motion intensity defines how the motion vector texture should influence the blending. It's needed at runtime because the Motion Vector texture is normalised and needs to be transformed before being used. By default, the motion intensity is in the file name and encoded in the blue and alpha channels. We only keep track of the fractional part of the intensity number (with an intensity of **0.0371** the file name would be 8x8_Explosion_MotionVectors_Intensity-**0371**).

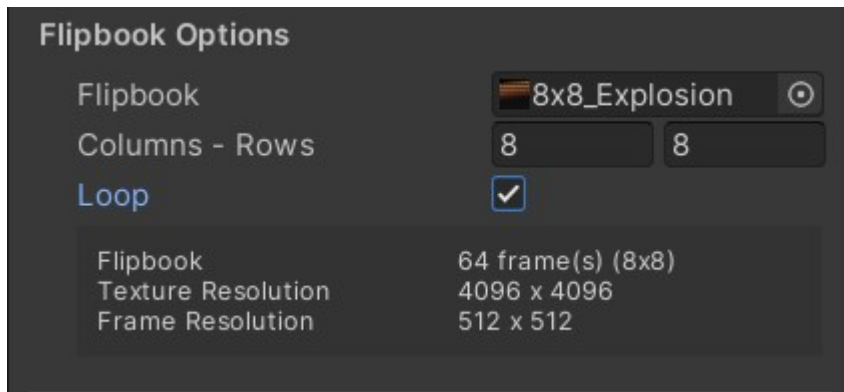
Storing the motion intensity in two places allows it to have a motion vectors texture that can be used in any pipeline.

If you choose to name the texture differently, you can always get the motion texture by right clicking **Tuatar** > **Copy Motion Intensity into clipboard**.



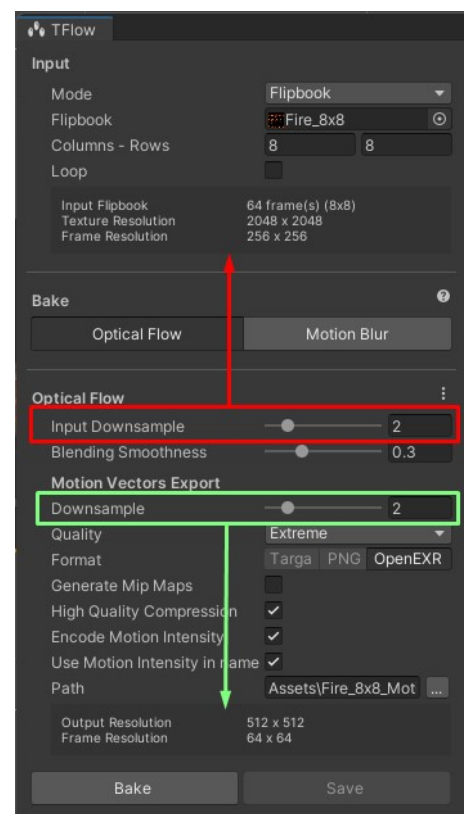
Looping flipbooks

If your flipbook animation is supposed to loop, enable the Loop checkbox so that the last frame blends with the first frame. Otherwise, the last frame fades out.



Downsampling

- **Input Downsample (red)**
Increase this value to reduce the input size before generating motion vectors. Higher values yield smoother and faster results but reduce fidelity. When baking is slow, increase this value. Baking time will slow down as the input texture size gets bigger.
- **Output Downsample (green)**
Increase this value as much as possible to get the smallest motion vectors texture and thus save runtime performance. Don't increase it if the blending looks worse. Changing this doesn't affect the baking speed.



Universal RP

Extract Universal RP package located in **TFlow/Runtime/Examples** folder. Open Universal RP scene located in **TFlow/Runtime/Examples/Universal RP** folder.

1. Particle System Shaders

We have added optical flow behaviours in our own version of Unity Universal RP particle shaders. You can use them in your materials by selecting one of these:

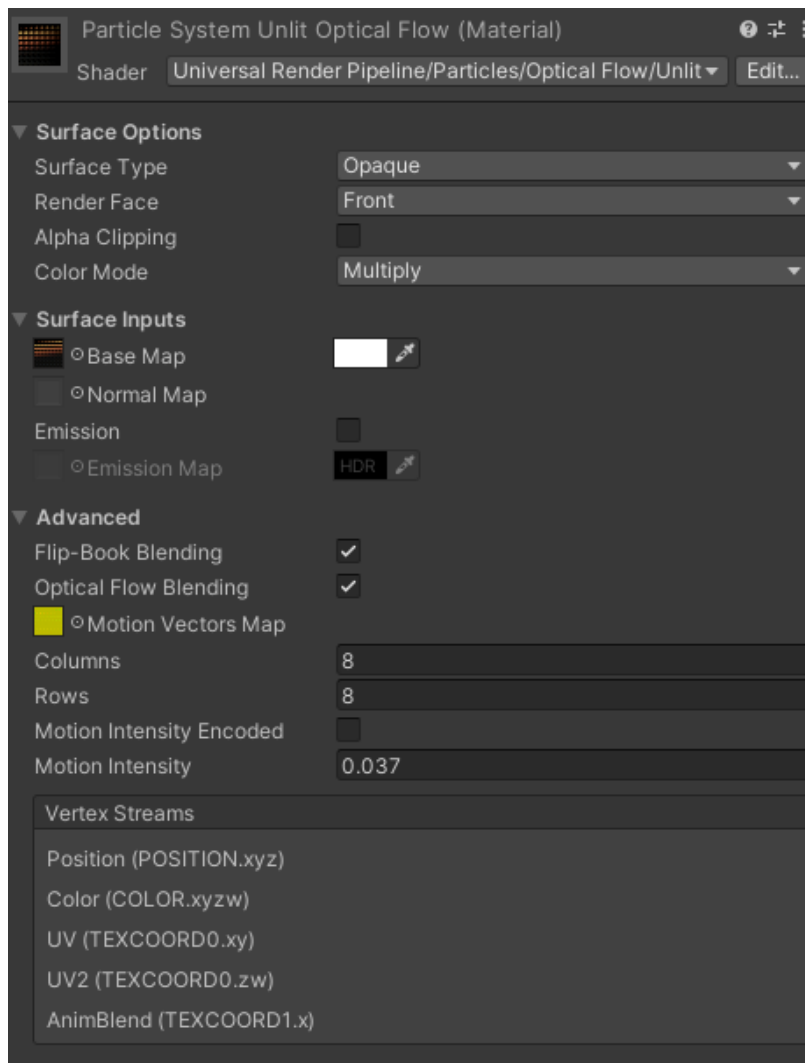
Universal Render Pipeline/Particles/Optical Flow/Lit

Universal Render Pipeline/Particles/Optical Flow/Simple Lit

Universal Render Pipeline/Particles/Optical Flow/Unlit

To enable optical flow behaviours, you must enable **Flip-Book Blending** and **Optical Flow Blending** and fill the properties correctly.

1. **Motion Vector** - The texture computed using TFlow baker.
2. **Columns and Rows** - The related frame count of your flipbook.
3. **Motion Intensity Encoded** - Did you encode the intensity in the Motion Vector texture ?
4. **Motion Intensity** - If the value is not encoded in the texture then use the value contained in the texture file name (Ex: 8x8_Explosion_MotionVectors_Intensity-0371 fill 0.0371). You can also get the value in the clipboard by right clicking on the texture (see [Motion Intensity](#) section).



2. Example Shader

We provide a shader example made with **Shader Graph** to illustrate the usage of the custom nodes. This shader can be selected in **Tuatara/OpticalFlow**. For more detailed explanation of shader creation with optical flow behaviours using **Shader Graph**, see [Shader Graph](#) section.

3. Shader Functions

We provide a set of functions that you can use in custom shaders or in any node based shader authoring tool. For more information about those functions, see [Shader Functions](#) section.

4. VFX Graph

We provide a VFX Graph example. By using the VFX Graph without any custom shader, optical flow works out of the box with Uv Mode set to Flipbook Motion Blend. For more information about these functions, see [VFX Graph section](#).

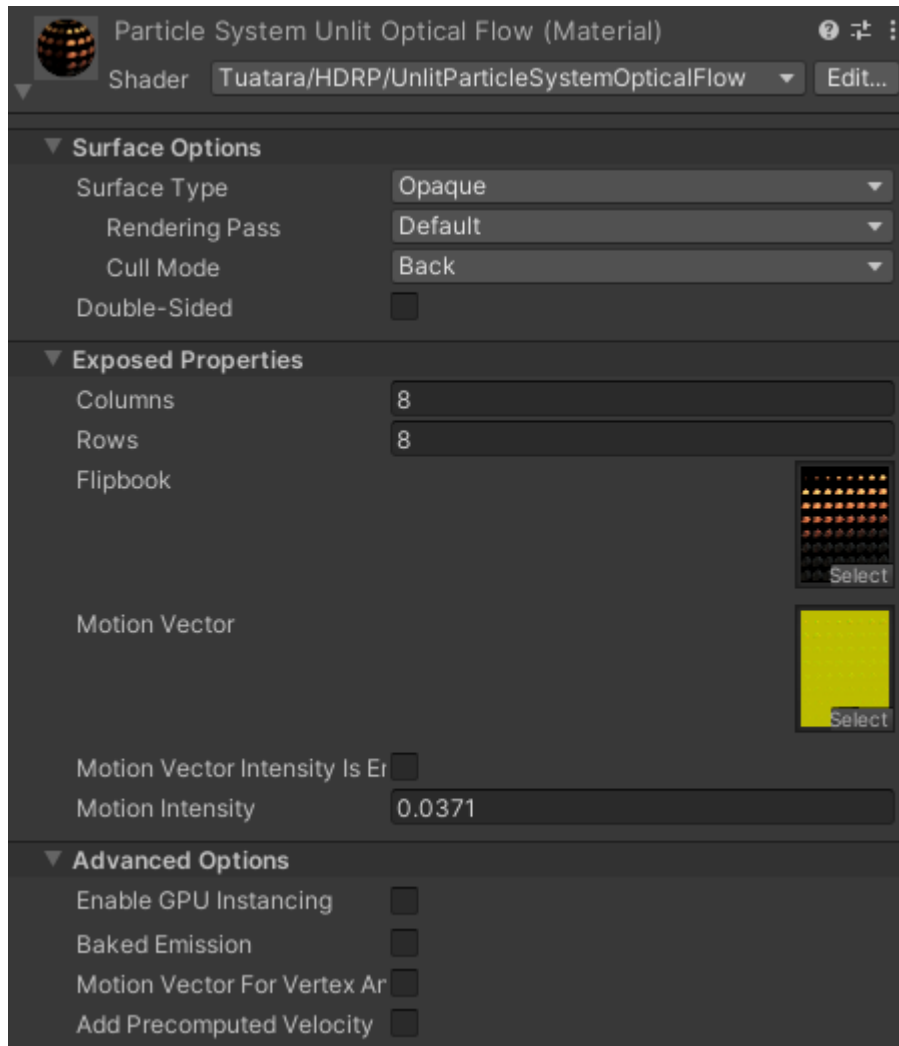
High Definition RP

Extract High Definition RP package located in **TFlow/Runtime/Examples** folder. Open High Definition RP scene located in **TFlow/Runtime/Examples/High Definition RP** folder.

1. Particle System Shader

We provide an unlit particle system compatible shader, it can be selected in **Tuatara/HDRP/UnlitParticleSystemOpticalFlow**. Here are the properties associated with the optical flow behaviours.

1. **Columns and Rows** - The related frame count of your flipbook.
2. **Motion Vector Intensity Is Encoded** - Did you encode the intensity in the Motion Vector texture ?
3. **Motion Intensity** - If the value is not encoded in the texture then use the value contained in the texture file name (Ex: 8x8_Explosion_MotionVectors_Intensity-**0371** fill **0.0371**). You can also get the value in the clipboard by right clicking on the texture (see [Motion Intensity](#) section).



2. Example Shader

We provide a shader example made with **Shader Graph** to illustrate the usage of the custom nodes. This shader can be selected in **Tuatara/HDRP/UnlitOpticalFlow**. For more detailed explanation of shader creation with optical flow behaviours using **Shader Graph**, see [Shader Graph](#) section.

3. Shader Functions

We provide a set of functions that you can use in custom shaders or in any node based shader authoring tool. For more information about those functions, see [Shader Functions](#) section.

4. VFX Graph

We provide a VFX Graph example. By using the VFX Graph without any custom shader, optical flow works out of the box with **Uv Mode** set to **Flipbook Motion Blend**. For more information about those functions, see [VFX Graph](#) section.

Amplify Shader Editor

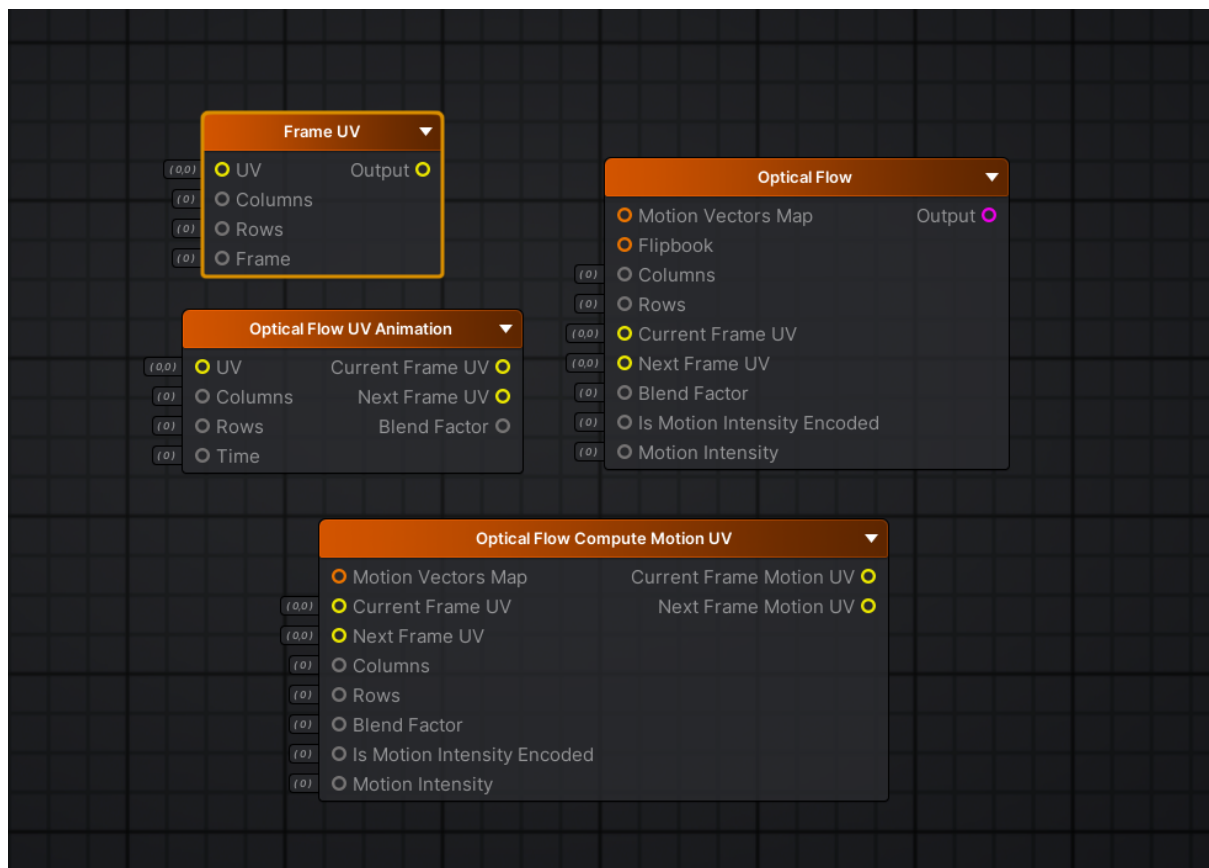
Extract **AmplifyShaderEditor** package located in **TFlow/Runtime/Examples** folder.

Open **AmplifyShaderEditor** scene located in

TFlow/Runtime/Examples/AmplifyShaderEditor folder.

You will find a shader example and several shader functions. These shader functions have several purposes.

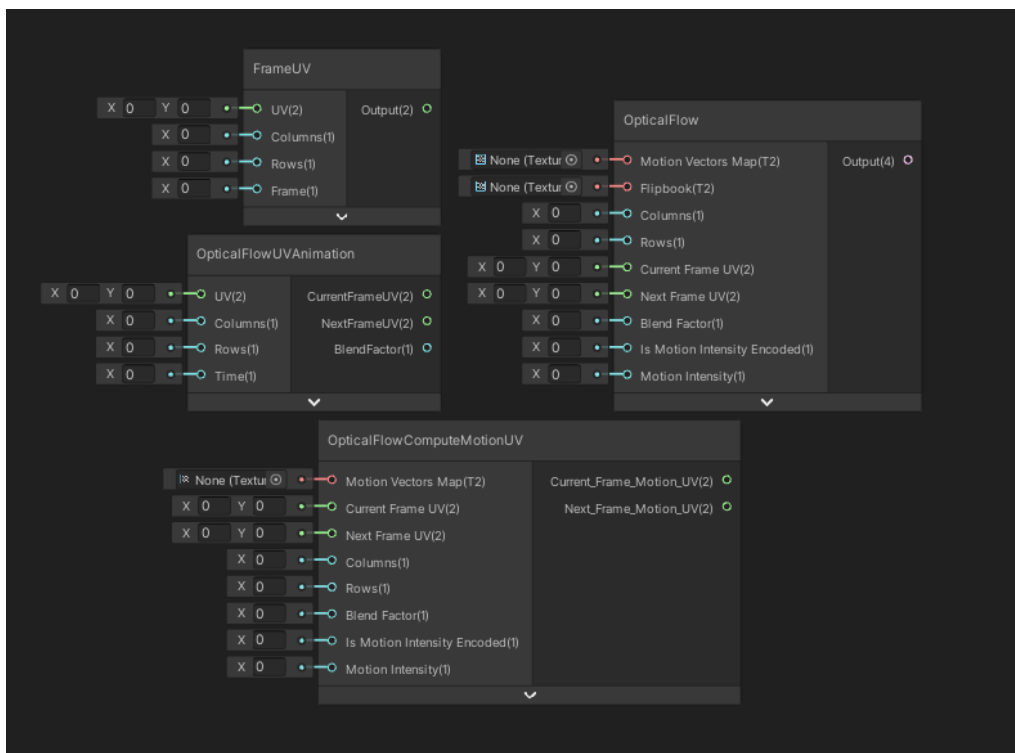
- **Frame UV** - Outputs a frame subUV based on flipbook properties (columns and rows) and current frame. This can be used to drive a flipbook based animation.
 - **Optical Flow UV Animation** - Outputs current frame subUV, next frame subUV and a blending factor between current and next frame. This can be used to drive a flipbook based animation with frame blending.
 - **Optical Flow Compute Motion UV** - Outputs current frame subUV and next frame subUV with taking motion vector and flipbook properties into account.
- Optical Flow - Outputs optical flow blending result of the current and next frames from flipbook and its properties.



Shader Graph

The **Shader Graph** examples provided in the **High Definition RP** and **Universal RP** packages are using several Sub Graphs to compute optical flow blending. Those Sub Graphs are located in the **TFlow/Runtime/Examples/Shader Graph**.

- **Frame UV** - Outputs a frame subUV based on flipbook properties (columns and rows) and current frame. This can be used to drive a flipbook based animation.
 - **Optical Flow UV Animation** - Outputs current frame subUV, next frame subUV and a blending factor between current and next frame. This can be used to drive a flipbook based animation with frame blending.
 - **Optical Flow Compute Motion UV** - Outputs current frame subUV and next frame subUV with taking motion vector and flipbook properties into account.
- Optical Flow** - Outputs optical flow blending result of the current and next frames from flipbook and its properties.



Shader Functions

This is the core of the package. With these functions, you'll be able to handle optical flow in any type of shader or pipeline. You can find all the functions in the **OpticalFlowCommon.hlsli** file located in the **TFlow/Runtime/Shaders** folder.

- **FrameUV_float** - Outputs a frame subUV based on flipbook properties (columns and rows) and current frame. This can be used to drive a flipbook based animation.

- **OpticalFlowUVAnimation_float** - Outputs current frame subUV, next frame subUV and a blending factor between current and next frame. This can be used to drive a flipbook based animation with frame blending.
- **OpticalFlowComputeMotionUV_float** - Outputs current frame subUV and next frame subUV with taking motion vector and flipbook properties into account.

Here is some pseudo code to use these functions.

```
float2 uv = texcoord0;
float time = 14.5;
float columns = 8;
float rows = 8;

float2 uvCurrentFrame, uvNextFrame;
float blendFactor;
OpticalFlowUVAnimation_float(uv, columns, rows, uvCurrentFrame, uvNextFrame,
blendFactor);

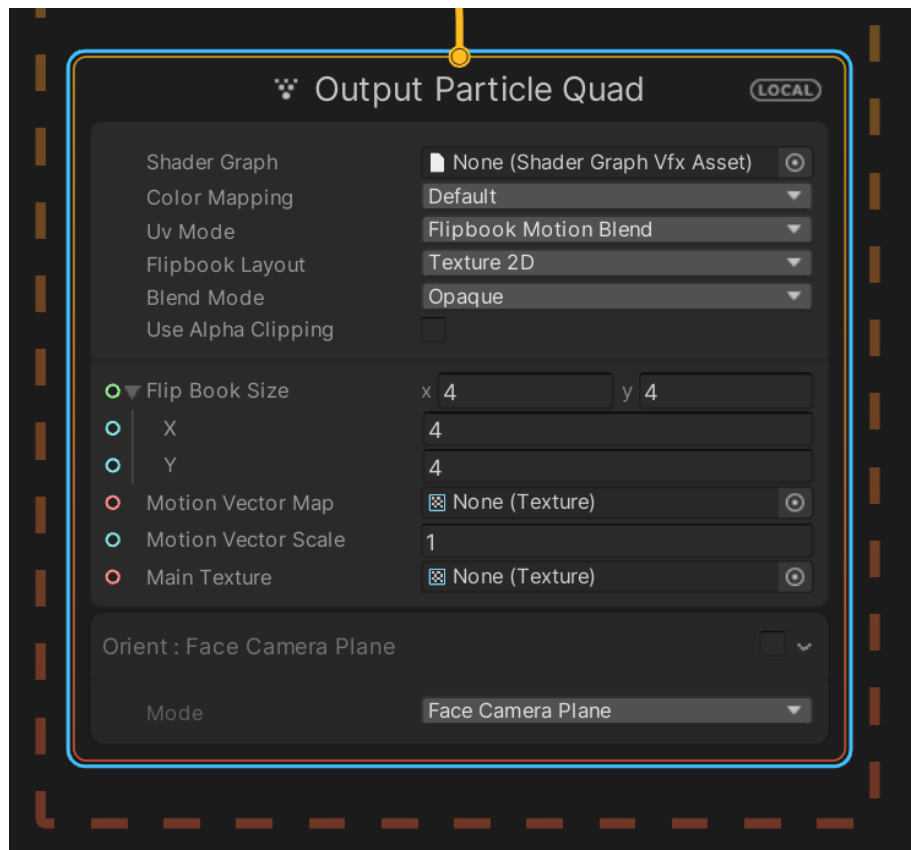
float4 currentMotionVectors = tex2D(motionVectorsMap, uvCurrentFrame);
float2 nextMotionVectors = tex2D(motionVectorsMap, uvNextFrame);
float isEncoded = 0.0;
float intensity = 0.0371;

float2 currentFrameMotionUV, nextFrameMotionUV;
OpticalFlowComputeMotionUV_float(      uvCurrentFrame, uvNextFrame,
currentMotionVectors, nextMotionVectors,
isEncoded, intensity,
columns, rows, blendFactor,
currentFrameMotionUV, nextFrameMotionUV);

float4 currentMotionVectors = tex2D(flipbookTex, currentFrameMotionUV);
float4 nextMotionVectors = tex2D(flipbookTex, nextFrameMotionUV);
float4 color = lerp(currentMotionVectors, nextMotionVectors, blendFactor);
```

VFX Graph

We provide simple VFX Graph examples in the **High Definition RP** and **Universal RP** packages that are using default shader output nodes. To enable optical flow blending behaviours, you need to set **Uv Mode** as **Flipbook Motion Blend** and provide the computed motion vector map. You can set the **The Motion Vector Scale** manually or get its value using the texture just like in the examples.



Resources

- [Tuatara](#)
- [Free VFX Image Sequences & Flipbooks](#)

Changelog

1.2.5 - 2022-08-26

Fixed

- Shader compilation error in Legacy for the function DecodeFloatRG

1.2.4 - 2022-03-21

Added

- Warning when using Gamma color space and OpenEXR format

1.2.3 - 2022-03-04

Fixed

- Shader compilation error in Legacy

1.2.2 - 2022-03-03

Fixed

- No Motion Vector when using uncompressed EXR
- Export Motion Blur in the same format as the input

1.2.1 - 2022-03-01

Fixed

- Error when reloading window

1.2.0 - 2022-02-24

Added

- Motion blur flipbook baking

Changed

- User interface refactor

Fixed

- Crash when using a compressed EXR input

1.1.1 - 2021-11-08

Fixed

- TGA format now work on all platforms
- Don't encode EXR images as TGA
- Don't encode TGA images as float

1.1.0 - 2021-10-28

Added

- Load a sequence of images or a flipbook
- Export a sequence of images or a flipbook

Changed

- Improved output settings
- Wording and typos

1.0.0 - 2021-10-01

Initial tool release.